

RACTF 2020 Crypto - 03

deuterium — 2020-06-09

Canonical HTML: <https://deut-erium.github.io/WriteUps/2020/ractf/crypto/03/2020-06-09-RACTF-2020-03.html>

Abstract

CTF writeup: Letter frequencies rat out a transposition in RACTF 2020's 03. The plaintext took a five-rail fence detour.

03

We are now provided with another cipher, and like 02 and 01, we dont know what cipher is it.

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GTHTI UHWSE ESLDL MUSDO RIROA
SRGER TAETL VSSAT OAONT EGESN
EOTNT GWPWI AFLAE OAIYA EAWTT
SMENO LTOTO AIASH RKLIC EEEYO
ESSUR NDBTA TNOES CMORI CEEIW
GDECO HSGEN UISIY EAERE YBEHT
LSRLN ADFHR SNTRM SUACU TTNRH
EWDHA EEIIO RHEND PFOLT TGHSC
DULWT NSNEO IHREG EWDEU IUEMC
APIOI VORFT USTGP OOA0E HEOER
BOEPB DH0BA BEATT ENESE WTBTK
KEIED ICTIR EPOTE LLEN0 EEPIO
IAAMC ONONY OEHEN ESIMT LFEIV
CEHOR AHSET ETENL EHAPS TRRWE
ISAVR HVGTL BPERI TOKER AIIP0
HNIIC ONIAP BSMMF HAYST UDLYM
NONPA REBTH MLOEH NRTEU ITOCY
GSSIE VOEMR ODTEI IEENI CUOFS
WFUMS TAHSP PCILD 00YUE ENBCE
IAEVO TAEKG FSEAH DLCLE PNTIC
CNPEE TNOLL AITME EOTCH RMRIT
ANANH LWTOU EOECA AHUTO BTRSA
UC
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The first thing you should probably try while working with simple ciphers is calculating the index of coincidence, for the given ciphertext it turns out to be 0.06618, pretty close to that of English plaintext at 0.0667. A low IC(around or lower than 0.04) indicates a flatter probability distribution(that is, random text), while a high IC(around or higher than 0.06) indicates either a transposition cipher or a monoalphabetic substitution cipher. The next thing which you can try to be sure is checking letter frequency.

Heading to frequency analysis section of dcode.fr —Letter—Count—Percentage—

—:—:—:—:—:— —E— 80×—14.49%— —T— 50×—9.06%— —O— 48×—8.7% — —I—
40×—7.25%— —A— 39×—7.07%— —S— 34×—6.16%— —N— 31×—5.62%— —R—

29×—5.25%— —H— 27×—4.89%— —L— 23×—4.17%— —C— 21×—3.8% — —U—
18×—3.26%— —P— 16×—2.9% — —D— 15×—2.72%— —M— 15×—2.72%— —G—
12×—2.17%— —W— 12×—2.17%— —B— 12×—2.17%— —F— 9× —1.63%— —Y— 9×
—1.63%— —V— 7× —1.27%— —K— 5× —0.91%—

Comparing the above distribution, one could easily tell that no substitution is involved (hopefully), which means the given thing would be transposition cipher.

Thinking of various transposition ciphers, the first one which comes in mind railfence cipher. Once again, I am asking for help from dcode.fr, and it does help (with removed punctuations and spaces option) to give out plaintext

```
AGENTGREATWORKWITHTHEFIRSTMESSAGEINEVERWOULDHAVETHOUGHTITWOULDBEASIMPLECAESARCIPHERITL  
OOKSLIKEEVIL  
CORPAREDEFINITELYUPTOSOMETHINGSUSPICIOUSROCCOANDDONNIEBOTHAPPEARTOBESENIORMEMBERSOFTHE  
COMPANYBOARD  
SO THIS COULD BE REALLY BIG WE MANAGED TO INTERCEPT ANOTHER MESSAGE BETWEEN THEM BUT IT LOOKS LIKE THEY V  
E INCREASED  
THEIR SECURITY A BIT MORE THE CRYPTOLOGISTS ARE STILL ON LEAVE AND NONE OF THEM ARE RESPONDING TO THEIR E  
MAIL I ASSUME  
WE CAN CONTINUE TO COUNT ON YOU FOR THESE WHEN WE FOUND THE MESSAGE IT CAME WITH A SLIP OF PAPER WHICH I V  
E INCLUDED A PHOTO  
OF FOR YOU ALL THE BEST AGENT B THE SECRET CODE IS ANUALL LEAVE WITHOUT ANY SPACES
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Key Railfence size 5, offset 3, from bottom

Putting correct spaces, we can read

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AGENT GREAT WORK WITH THE FIRST MESSAGE I NEVER WOULD HAVE THOUGHT IT WOULD  
BE A SIMPLE CAESAR CIPHER IT LOOKS LIKE EVIL CORP ARE DEFINITELY UP TO  
SOMETHING SUSPICIOUS ROCCO AND DONNIE BOTH APPEAR TO BE SENIOR MEMBERS OF  
THE COMPANY BOARD SO THIS COULD BE REALLY BIG WE MANAGED TO INTERCEPT  
ANOTHER MESSAGE BETWEEN THE MBUT IT LOOKS LIKE THEY VE INCREASED THEIR  
SECURITY A BIT MORE THE CRYPTOLOGISTS ARE STILL ON LEAVE AND NONE OF THEM  
ARE RESPONDING TO THEIR EMAIL I ASSUME WE CAN CONTINUE TO COUNT ON YOU FOR  
THESE WHEN WE FOUND THE MESSAGE IT CAME WITH A SLIP OF PAPER WHICH I VE  
INCLUDED A PHOTO OF FOR YOU ALL THE BEST AGENT B THE SECRET CODE IS  
ANUALL LEAVE WITHOUT ANY SPACES
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And **ANUALL LEAVE** is finally the secret code which we seek

Check this video by @hyperreality for a better analysis

| Interactive controls are omitted from this PDF. See the canonical HTML version.